

Short answer: **Nahi. Ye flow sirf bahut se image classification projects (CNN-based) ke liye common hai, sab AI/ML projects ke liye nahi.**

Har type ke project ka apna workflow hota hai.

1. CNN Image Classification (Tumhara project)

Ye flow bilkul sahi hai.

Dataset Load



Data Augmentation (Optional)



Rescaling / Normalization



CNN Layers



Flatten / GlobalAveragePooling



Dense Layer



Output Layer



Compile



Train



Evaluate



Predict

2. Transfer Learning (MobileNet, ResNet, EfficientNet)

Dataset Load



Data Augmentation



Preprocessing



Pretrained Model



GlobalAveragePooling



Dense Layer



Output Layer



Compile



Train



Fine-tune (Optional)



Evaluate



Predict

Yahan Conv2D layers manually nahi likhte, kyunki pretrained model use hota hai.

3. Tabular Machine Learning (House Price, Fraud Detection)

Load Data



Clean Data



Handle Missing Values



Encode Categorical Data



Feature Engineering



Train-Test Split



Train Model



Evaluate



Predict



Save Model

Yahan CNN ya Flatten ka use nahi hota.

4. NLP (Text Classification)

Load Text Data



Text Cleaning



Tokenization



Padding



Embedding



LSTM / GRU / Transformer



Dense



Output



Compile



Train



Evaluate

5. Object Detection (YOLO)

Load Images + Labels



Data Augmentation



Preprocessing



YOLO Model



Train



Evaluate (mAP)



Detect Objects

6. Segmentation (U-Net)

Load Images + Masks



Augmentation



Normalization



Encoder



Decoder



Segmentation Mask



Compile



Train



Evaluate

Agar sirf TensorFlow/Keras me model banana ho, to ek generic template yaad rakh sakte ho:

1. Import Libraries



2. Load Dataset



3. Preprocess Data



4. Build Model



5. Compile Model



6. Train Model



7. Evaluate Model



8. Predict



9. Save Model

Ye template **90% TensorFlow/Keras projects** me kisi na kisi form me use hota hai. Sirf **Step 4 (Build Model)** project ke type ke hisaab se badal jata hai—CNN, ANN, LSTM, Transformer, U-Net, YOLO, etc.

Exam ya interview ke liye bhi ye generic flow yaad rakhna sabse useful hai.